

Orthogonal Matrices

A square matrix A over the field R is called Orthogonal if $A^t A = I = A A^t$

OR $A^t = A^{-1}$

A^t is inverse of A
 $\therefore A^t A = I$

From the def it follows that

- A is orthogonal iff A^{-1} is orthogonal
- If A is orthogonal then A is symmetric iff $A^2 = I$
- If A & B are orthogonal then A^t and B^t are also orthogonal.
- If A is orthogonal then the col vectors are of unit length and their dot products are zero i.e. Columns of A form an orthonormal set.
- If A is orthogonal then the rows of A form an orthonormal set.
- A is orthogonal iff A^t is orthogonal.

Theorem The following conditions for a square Matrix A are equivalent:

- A is orthogonal
- The rows of A form an orthonormal set.
- The col of A form an orthonormal set.

Proof Let $A = [a_{ij}]_{n \times n}$ be an Orthogonal Matrix i.e. $AA^t = I = A^t A$

Let R_i & C_j denote the i^{th} row & j^{th} col of A where $1 \leq i \leq n$

Let $C_{ij} = (i, j)$ element of AA^t

= Sum of products of the corresponding elements in the i^{th} row of A and j^{th} col of A^t .

= Sum of products of the corresponding elements in the i^{th} row of A and j^{th} row of A

$$= a_{i1}a_{j1} + a_{i2}a_{j2} + \dots + a_{in}a_{jn}$$

$$C_{ij} = \langle R_i, R_j \rangle \quad \text{--- (1)}$$

Matrix $[C_{ij}]$ (I.P of two rows)

Now $[C_{ij}] = AA^t$

$$[C_{ij}] = I$$

$$\therefore \langle R_i, R_j \rangle = C_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$$\therefore C_{ij} = \langle R_i, R_j \rangle$$

$$\therefore \langle R_i, R_j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$\therefore A$ is orthogonal so $AA^t = I$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix} \begin{matrix} \text{when } i=j \\ c_{ij}=1 \\ \text{when } i \neq j \\ c_{ij}=0 \end{matrix}$$

$\Rightarrow$ Rows of A are orthonormal
 $\Rightarrow$ (i) = (ii) (from statement & theorem)

Conversely if rows of A form an orthonormal set, then

$$\langle R_i, R_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\therefore \langle R_i, R_j \rangle = C_{ij}$$

$$\therefore C_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

Matrix $\Rightarrow [C_{ij}] = I$

$$\therefore AA^t = [C_{ij}] \therefore AA^t = I \Rightarrow A \text{ is orthogonal Matrix}$$

So (i) $\Rightarrow$ (i') Hence (i) $\Leftrightarrow$ (ii)

Now Consider the Matrix $A^t A = [d_{ij}]$

$$d_{ij} = (i, j) \text{ element of } A^t A$$

$$= \text{Sum of the products of the corresponding elements in } i^{\text{th}} \text{ row of } A^t \text{ and } j^{\text{th}} \text{ col of } A.$$

$$= \text{Sum of the products of the corresponding elements in } i^{\text{th}} \text{ col of } A \text{ and } j^{\text{th}} \text{ col of } A$$

$$= a_{i1}a_{j1} + a_{i2}a_{j2} + \dots + a_{in}a_{jn}$$

$$= \text{Inner product of } i^{\text{th}} \text{ and } j^{\text{th}} \text{ col matrices}$$

$$\begin{aligned} i^{\text{th}} \text{ row of } A^t &= (a_{i1}, a_{i2}, \dots, a_{in}) \\ i^{\text{th}} \text{ col of } A &= \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{pmatrix} \\ j^{\text{th}} \text{ col of } A &= \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{pmatrix} \end{aligned}$$

$$d_{ij} = \langle C_i, C_j \rangle$$

$$[d_{ij}] = A^t A = I$$

$$\therefore A \text{ is orthogonal Matrix}$$

$$\Rightarrow d_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\therefore d_{ij} = \langle C_i, C_j \rangle$$

$$\therefore \langle C_i, C_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\Rightarrow \text{cols } C_i \text{ and } C_j \text{ are orthonormal}$$

$$\Rightarrow (i) \Rightarrow (iii)$$

Conversely if columns of A form orthonormal set, then

$$\langle C_i, C_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\therefore \langle C_i, C_j \rangle = d_{ij} \therefore d_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\Rightarrow [d_{ij}] = I$$

$$\therefore [d_{ij}] = A^t A \therefore A^t A = I \Rightarrow A \text{ is orthogonal Matrix}$$

$$\text{So (iii) } \Rightarrow (i) \text{ Hence (i) } \Leftrightarrow (iii)$$

$$\text{Now (i) } \Leftrightarrow (ii) \text{ + (ii) } \Leftrightarrow (iii) \Rightarrow (ii) \Leftrightarrow (iii)$$

$$\text{N.B. (ii) } \Leftrightarrow (iii) \text{ so } A \text{ is orthogonal iff } A^t \text{ is orthogonal}$$

Example 13 Show that the rows (columns) of the matrix

$$A = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ form an orthonormal set.}$$

Sol To show that the rows (columns) of the matrix A form an orthonormal set we have only to show that A is orthogonal (Th 7.12 on page 277)
(If A is orthogonal then $AA^t = I$)

$$\begin{aligned} \therefore AA^t &= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta + 0 & \cos \theta \sin \theta - \sin \theta \cos \theta + 0 & 0 + 0 + 0 \\ \sin \theta \cos \theta - \cos \theta \sin \theta + 0 & \sin^2 \theta + \cos^2 \theta + 0 & 0 + 0 + 0 \\ 0 + 0 + 0 & 0 + 0 + 0 & 0 + 0 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Thus A is orthogonal
So Rows of A form an orthonormal set.

Example 14 Find an orthogonal matrix A whose first row is $\left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

Sol First Method Let $R_1 = \left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

A vector (row) orthogonal to R_1 is $\star R_2 = \begin{vmatrix} e_1 & e_2 \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{2}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix}$

$$\begin{aligned} &= e_1 \left(\frac{2}{3}\right) - e_2 \left(\frac{1}{3}\right) + e_2 \left(\frac{2}{3}\right) - e_3 \left(\frac{2}{3}\right) + e_3 \left(\frac{1}{3}\right) - e_1 \left(\frac{2}{3}\right) \\ &= e_2 \left(\frac{2}{3} - \frac{1}{3}\right) + e_3 \left(-\frac{2}{3} + \frac{1}{3}\right) = e_2 \left(\frac{1}{3}\right) + e_3 \left(-\frac{1}{3}\right) \\ \star R_2 &= \left(0, \frac{1}{3}, -\frac{1}{3}\right) \end{aligned}$$

$$R_2 = \frac{\star R_2}{\|\star R_2\|} = \frac{(0, \frac{1}{3}, -\frac{1}{3})}{\frac{1}{3}\sqrt{2}} = \frac{3}{\sqrt{2}} \left(0, \frac{1}{3}, -\frac{1}{3}\right) = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

A vector (row) orthogonal to R_1 & $\star R_2$ is $\star R_3 = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{3} & -\frac{1}{3} \end{vmatrix} = e_1 \left(-\frac{2}{9} - \frac{2}{9}\right) - e_2 \left(-\frac{1}{9}\right) + e_3 \left(\frac{1}{9}\right)$

$$\begin{aligned} &= -\frac{4}{9} e_1 + \frac{1}{9} e_2 + \frac{1}{9} e_3 \\ &= \left(-\frac{4}{9}, \frac{1}{9}, \frac{1}{9}\right) \end{aligned}$$

$$R_3 = \frac{\star R_3}{\|\star R_3\|} = \frac{\left(-\frac{4}{9}, \frac{1}{9}, \frac{1}{9}\right)}{\frac{1}{9}\sqrt{18}} = \frac{9}{3\sqrt{2}} \left(-\frac{4}{9}, \frac{1}{9}, \frac{1}{9}\right) = \left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}\right)$$

$$\therefore \text{Orthogonal Matrix } A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} \end{bmatrix}$$

(To check Find $A^t A$.
we get $A^t A = I$)

Example 14 2nd Method

7.2A-4

Q4

$$\text{Let } R_1 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right)$$

A vector orthogonal to R_1 is R_2^*

$$R_2^* = \begin{vmatrix} e_1 & e_2 \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{2}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix}$$

$$= e_1 \left(\frac{2}{3} \right) - e_2 \left(\frac{1}{3} \right) + e_2 \left(\frac{2}{3} \right) - e_3 \left(\frac{2}{3} \right) + e_3 \left(\frac{1}{3} \right) - e_1 \left(\frac{2}{3} \right)$$

$$R_2^* = e_2 \left(\frac{1}{3} \right) + e_3 \left(-\frac{1}{3} \right) = \left(0, \frac{1}{3}, -\frac{1}{3} \right)$$

$$\|R_2^*\| = \sqrt{0^2 + \frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3}$$

$$R_2 = \frac{R_2^*}{\|R_2^*\|} = \frac{\left(0, \frac{1}{3}, -\frac{1}{3} \right)}{\frac{\sqrt{2}}{3}} = \frac{3}{\sqrt{2}} \left(0, \frac{1}{3}, -\frac{1}{3} \right) = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

A vector orthogonal to R_1 & R_2 is R_3^*

$$R_3^* = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{vmatrix} = e_1 \left(-\frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} \right) - e_2 \left(-\frac{1}{3\sqrt{2}} \right) + e_3 \left(\frac{1}{3\sqrt{2}} \right)$$

$$= \left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right)$$

$$R_3 = \frac{R_3^*}{\|R_3^*\|} = \frac{\left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right)}{\frac{1}{\sqrt{2}}} = \left(-\frac{4}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

Hence orthogonal Matrix $A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} \end{bmatrix}$

Note

We can check our answer by $\langle R_1, R_2 \rangle = 0 = \langle R_1, R_3 \rangle$
 $0 = \langle R_2, R_3 \rangle$
 $\langle R_1, R_1 \rangle = \langle R_2, R_2 \rangle = 1 = \langle R_3, R_3 \rangle$

OR $AA^T = I$

3rd Method

Let $v_1 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right)$ & $w_2 = (x, y, z)$

Let w_2 is orthogonal to $v_1 \therefore \langle v_1, w_2 \rangle = 0$

$$\Rightarrow \frac{x}{3} + \frac{2y}{3} + \frac{2z}{3} = 0 \text{ or } x + 2y + 2z = 0 \text{ --- (I)}$$

If we take $x=0$ then $2y + 2z = 0 \Rightarrow z = -y$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ y \\ -y \end{bmatrix} = y \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \therefore w_2 = (x, y, z) = (0, 1, -1)$$

$$v_2 = \frac{w_2}{\|w_2\|} = \frac{(0, 1, -1)}{\sqrt{2}} = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

Now let $w_3 = (x, y, z)$ be orthogonal to v_1 & v_2

$$\therefore \langle v_1, w_3 \rangle = 0 \Rightarrow \frac{x}{3} + \frac{2y}{3} + \frac{2z}{3} = 0 \Rightarrow x + 2y + 2z = 0 \text{ --- (II)}$$

$$\langle v_2, w_3 \rangle = 0 \Rightarrow \frac{y}{\sqrt{2}} - \frac{z}{\sqrt{2}} = 0 \Rightarrow y - z = 0 \Rightarrow y = z \text{ Put in (II)}$$

$$\Rightarrow x = -4z \text{ from (II)}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4z \\ z \\ z \end{bmatrix} = z \begin{bmatrix} -4 \\ 1 \\ 1 \end{bmatrix} \therefore w_3 = (-4, 1, 1)$$

$$v_3 = \frac{w_3}{\|w_3\|} = \frac{(-4, 1, 1)}{\sqrt{18}} = \left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right) \text{ Hence } A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} \end{bmatrix}$$

$\checkmark \Rightarrow$ order of A is 3×3
 $\therefore A$ must be square.
 v_1 is first row of A
 v_2 is 2nd row of A
 (we may take $y=0$ or $z=0$)